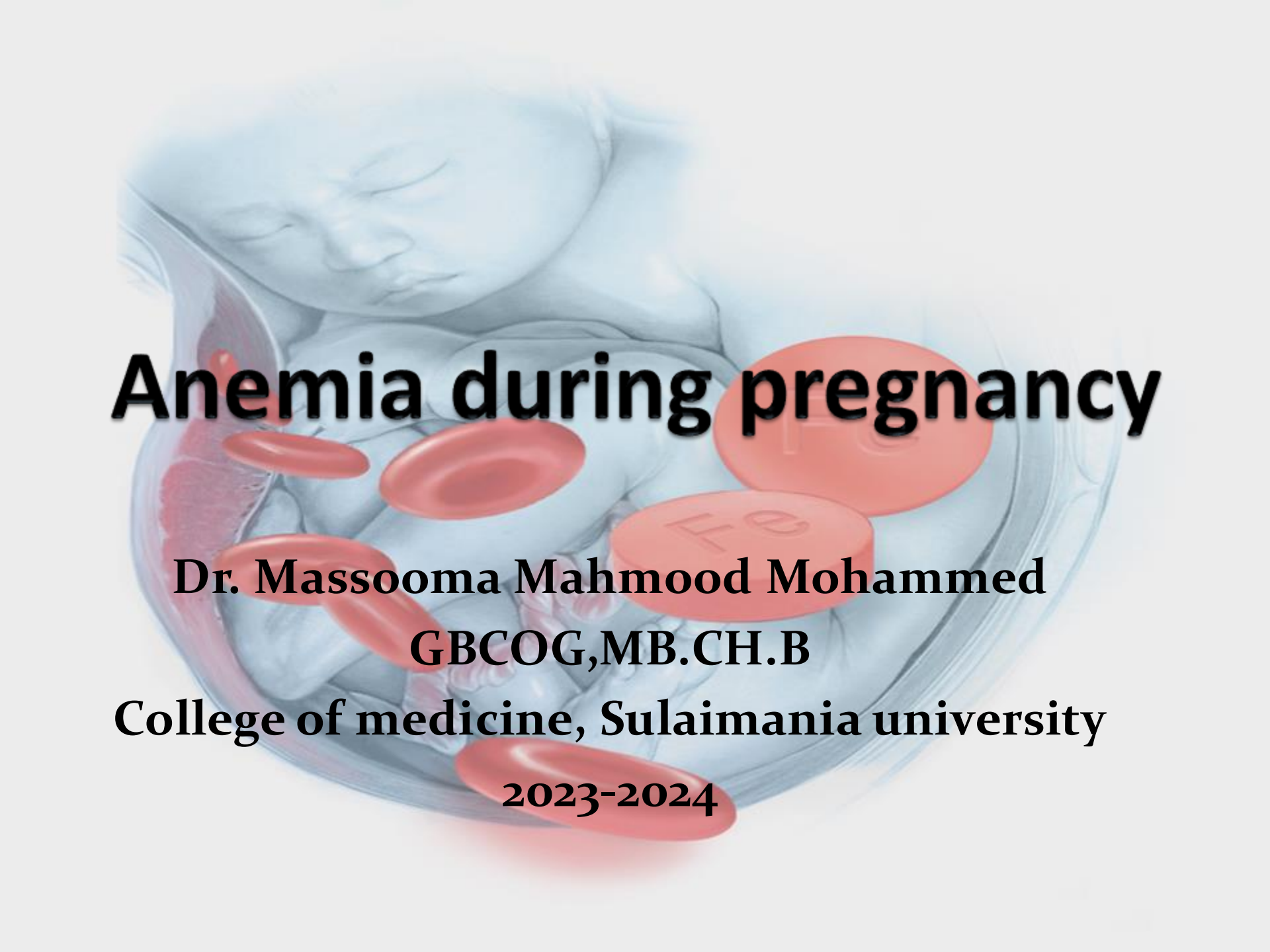




Anemia during pregnancy

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2023-2024

Learning Objectives

- Background information
- Diagnose anemia in pregnancy
- Learning the effect on mother & fetus
- Learning S/S in pregnancy
- Learning prevention of anemia
- Learning supplementation of oral iron during pregnancy
- Management of anemia during pregnancy

Physiological changes in pregnancy

- **Plasma volume** increase 50%
- **Red cell mass** increase 25%
- **Hb** , haematocrit ,RBC count fall due to haemodilution
- **(TIBC)**Total iron binding capacity increases.
- **Serum iron and ferritin** concentration decreases due to increased utilization. **Iron** requirements increase from **2.5 mg /day** in 1st trimester to **6.6mg/day** (totally 700-1400 mg)
- iron absorption moderately increase
- **folate** requirement increase
- no major change in **Vit. B12** store but the level decrease due to active transport to the fetus.

Background Information

- Commonest medical disorder in pregnancy
- Prevalence in USA is 2-4%
- Nutritional anemia (Fe deficiency) is commonest
- It is important contributor to maternal & perinatal morbidity & mortality as a direct or indirect cause

Types of Anemia During Pregnancy:

- **Iron-deficiency anemia:**
- Folate-deficiency anemia
- Vitamin B12 deficiency

Definition

- A condition where circulating levels of Hb are quantitatively or qualitatively lower than normal

Or a pathological condition in which the oxygen carrying capacity of RBC is insufficient to meet the body's needs.

- 1st trimester: Hb < 11 g/dL; Hct. < 33%
- 2nd trimester: Hb < 10.5 g/dL; Hct. < 32%
- 3rd trimester: Hb < 11 g/dL; Hct. < 33%

- **Severity of Anemia:**

- Mild 10.0-10.9 gm%
- Moderate 7-9.9
- Severe <7
- Very Severe <4

Causes of Anemia in Pregnancy

- Nutritional / Iron deficiency anemia
- Pre-pregnancy poor nutrition very important
- Besides Iron, folate and B12 deficiency also important
- Chronic blood loss due to parasitic infections – Hookworm & malaria
- Multiparity
- Multiple pregnancy
- Acute blood loss in APH, PPH
- Recurrent infections (UTI) - anemia due to impaired erythropoiesis
- Hemolytic anemia in PIH
- Hemoglobinopathies like Thalassemia, sickle cell anemia
- Aplastic anemia is rare
- Morning sickness
- Very young mother

Clinical presentation

- Depends on severity of anemia
- **High risk women** – adolescent, multiparous, multiple pregnancy, lower socio economic status
- **Mild anemic - asymptomatic**
- **Symptoms** – pallor, weakness, fatigue, dyspnoea, palpitation, swelling over feet & body
- **Signs** – pallor, facial puffiness, raised JVP, tachycardia, tachypnea, crepts in lung bases, hepato-splenomegaly, pitting oedema over abdominal wall & legs
- Haemic murmur, cardiac failure
- Glossitis, stomatitis, chelosis, brittle hair

Symptoms of **Anemia**

Red = In severe
anemia

Eyes
- Yellowing

Skin
- Paleness
- Coldness
- Yellowing

Respiratory
- Shortness of breath

Muscular
- Weakness

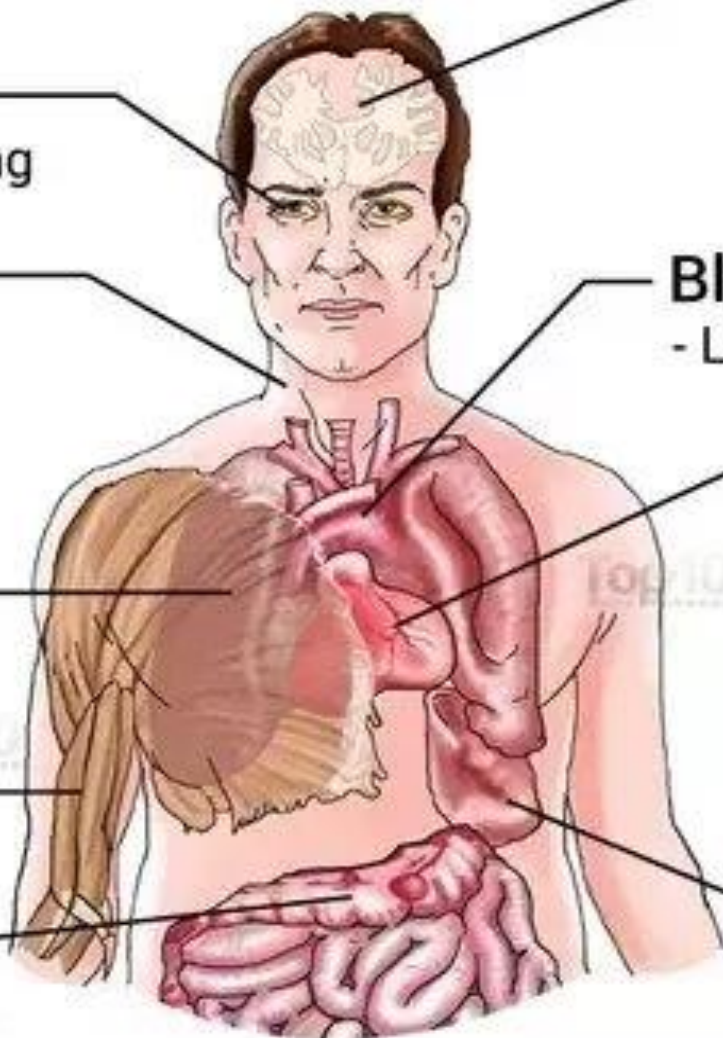
Intestinal
- Changed stool color

Central
- Fatigue
- Dizziness
- **Fainting**

Blood vessels
- Low blood pressure

Heart
- Palpitations
- Rapid heart rate
- **Chest pain**
- **Angina**
- **Heart attack**

Spleen
- Enlargement



Effect of Anemia on Pregnancy & Mother

- Higher incidence of pregnancy complications
 - PET, abruptio placentae, preterm labor
- Predisposed to infections like – UTI, puerperal sepsis
- Increased risk to PPH
- Subinvolution of uterus
- Lactation failure
- Maternal mortality – due to
 - CHF,
 - Cerebral anoxia,
 - Sepsis,
 - Thrombo-embolism

Effect of Anemia on Fetus & Neonate

- Higher incidence of abortions, preterm birth, IUGR
- IUD
- Low APGAR at birth
- Neonate more susceptible for anemia & infections
- Higher Perinatal morbidity & mortality
- Anemic infant with cognitive & affective dysfunction

Effects of Anemia on Pregnancy

Fetus

- Neural Tube defects (esp. folate def.)
- Miscarriage
- OIUGR/ Low birth weight
- Prematurity
- Anemia in infancy
- IUFD

Mother

- Susceptibility to infection
- Heart decompensation and Heart failure
- Preterm labour and Preterm delivery
- Post-partum hemorrhage
- Mental lassitude & Loss of working hours
- Death



Work Up of Pregnancy with Anemia

- Detailed H/o – age, parity, diet, chronic bleeding, worm infestation, malaria, race etc...
- Examination
 - Pallor
 - Glossitis
 - Splenomegaly – hemolytic anemia
 - Jaundice – hemolytic anemia
 - Purpura – bleeding disorder
 - Evidence of chronic disease – Renal , TB
 - Anasarca & signs of cardiac failure in severe cases

Investigations

- Severity of anemia – Hb & Haematocrit, at first visit, 28-30 weeks & 36 weeks
- Type of anemia – **GBP** microcytic, macrocytic, dimorphic, normocytic, hemolytic, pancytopenia
- Bone marrow activity – reticulocyte count (N 0.2-2%), higher bone marrow activity is seen in
 - hemolytic anemia
 - following acute blood loss
 - iron deficiency anemia on treatment
- **CBC: RBC count** – decreases in anemia (N 3.2 million/cu mm)
- **PCV** - < 32%, (N37-47%)
- **MCV** – low in Fe def. anemia, microcytic
- **MCH** - decreases
- **MCHC** – decreases, one of the most sensitive indices (N26-30%)

Investigations

- Cause of anemia – by various investigations

e.g.:

- **Serum Ferritin** – abnormal if < 20 ng/ml (N 40-160 ng/dl), assess iron stores
- **Serum Iron** – N 65-165 ug/dl, decreases in Fe def anemia
- **Serum Iron binding capacity** – 300-360 ug/dl, increases with severity of anemia
- **Percentage saturation of transferrin** – 35-50%, decreases to less than 20% in iron deficiency anemia
- Other examples???

Treatment for Iron Deficiency Anemia

- **Improving diet** rich in iron & fruits & leafy vegetables
- **Treat worm infections**, maintain general hygiene
- Food fortification with iron & genetic modification of food
- **Iron & folic acid** supplementation(By folate deficiency 1mg folate twice daily
- Iron absorption enhanced by citrous fruits, Vit C
- Avoid tea, coffee, Ca, phytates, phosphates, oxalates, egg, cereals with iron

Iron supplementation in Pregnancy

- 60 mg elemental iron & 400 ug of folic acid daily during pregnancy and 3 months there after
- In anemia therapeutic doses are 180-200 mg /d
- Route of administration depends on, severity of anemia, gestational age, compliance & tolerability of iron
- Various preparations – fumarate, gluconate, succinate, sulfate, ascorbate
- Carbonyl iron better tolerated
- Oral iron can have side effects like nausea, vomiting, gastritis, diarrhoea, constipation
- Iron sucrose for parenteral use
- Response - by increase in Hb level 1g/week
- Clinical symptoms improve

Indications for Blood Transfusion

- Severe anemia first seen after 36 weeks of pregnancy
- Anemia due to acute blood Loss – APH & PPH
- Associated Infection
- Patient not responding to oral or parenteral therapy
- Anemic & symptomatic pregnant women (dyspneic, with heart failure etc.) irrespective of gestational age

Hemoglobinopathies in Pregnancy

- During pregnancy, hemoglobinopathies, particularly sickle cell disease, Hb S-C disease, and beta- and alpha-thalassemia, can worsen maternal and perinatal outcomes.
- Anemia almost always becomes more severe as pregnancy progresses.
- Treatment of sickle cell disease during pregnancy is complex. Painful crises should be treated aggressively. Prophylactic exchange transfusions to keep Hb A at $\geq 60\%$ reduce risk of hemolytic crises and pulmonary complications, but they are not routinely recommended because they increase risk of transfusion reactions, hepatitis, HIV transmission, and blood group isoimmunization. Prophylactic transfusion does not appear to decrease perinatal risk.

Hemoglobinopathies in Pregnancy

- **Therapeutic:**
- **Transfusion is indicated for the following:**
- Symptomatic anemia
- Heart failure
- Severe bacterial infection
- Severe complications of labor and delivery (eg, bleeding, sepsis)
- **beta-Thalassemia** is similar to Hb S-C disease but is less common and more benign.
- **Alpha-Thalassemia** does not cause maternal morbidity, but if the fetus is homozygous, hydrops and fetal death occur during the 2nd or early 3rd trimester

Assessment

1- Pregnant woman is considered anemic when her Hb is below (unit gm/dl):

- A. 12 B. 11 C. 10 D. 9

2- Most common cause of anemia in pregnancy in middle east is :

- A. Nutritional anemia
- B. Parasitic anemia
- C. Aplastic anemia
- D. Thalassemia

3- Iron deficiency anemia can be diagnosed earliest by which laboratory test?

- A. Hb%
- B. Serum ferritin
- C. Serum iron
- D. RBC protoporphyrin

4- Which complication is not common in Pregnancy with anemia ?

- A. PIH
- B. Preterm labour
- C. GDM
- D. Puerperal sepsis

